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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving Questions

Course Code:	CSA0303	**Course Title:**	Data Structures
CO Assessed:	CO2	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	**Total Marks:**	100
CO2 Statement:	ADT array, Operations on arrays, Searching Algorithms, Linear Search, Binary Search, Suffix Arrays & LCP Arrays ADT, Linked List, Stack, Queue, Applications.		
Student Name:	N.Thejasree	**Reg. No.:**	192524438
Section / Batch:	Sloat C	**Date:**	

Code of Conduct

I __N.Thejasree(192524438)________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.Thejasree

Instructions

- This test contains 3 Analytical Problem Solving questions. Total: 100 Marks.
- When a student answer using a analytical problem, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are not permitted in the class. Take it as test and submit in the class.

Section / Topic	Focus	Q Nos	Marks
Stack Data Structures	Stack – top movements	1	30
Queue Data Structures	Circular Queue	2	40
Application of Stack	Infix to post-fix	3	40
		GRAND TOTAL:	100 Marks

CO2 - DP1 Analytical.

N. Thejasree

1925242138

AIDS

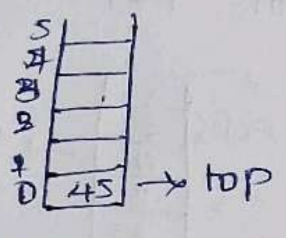
CSA0303 - Data structures

26/30

perform the following operation in LIFO data structure having an array size of 5

push(45)
push(22)
push(11)
push(77)
pop(),
push(44),
push(33),
pop(),
pop().
pop().

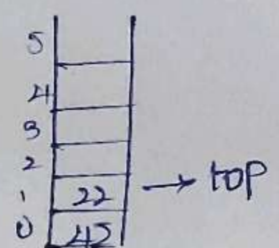
1. push(45) size = 5



if it is Empty
top = -1

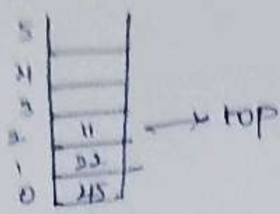
top = top + 1
top = 0 + 1
top = 1

push(22)



top = 1 + 1
top. = 2

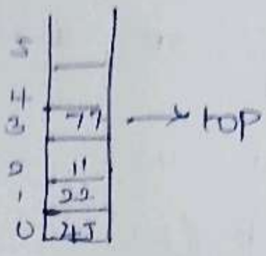
iii. push(11)



top = 2+1

top = 3

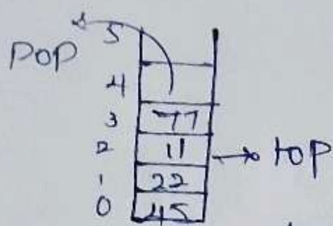
iv. push(77)



top = 3+1

top = 4

v. pop().

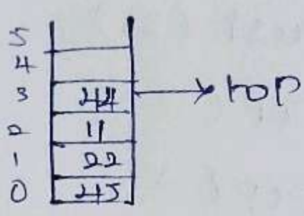


top = top -1

top = 3+1

top = 2

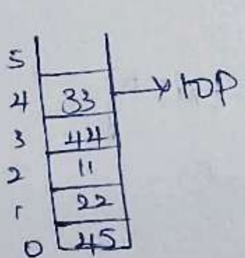
vi. push(44)



top = 3+1

top = 4

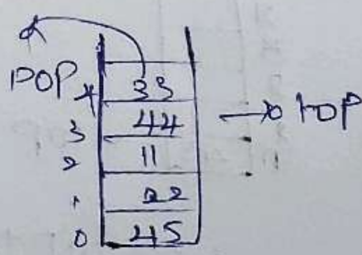
vii. push(33)



top = 4+1

top = 5

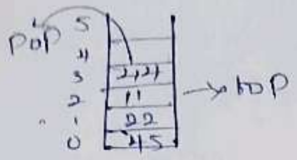
viii. POP( )



top = 4 -1

top = 3

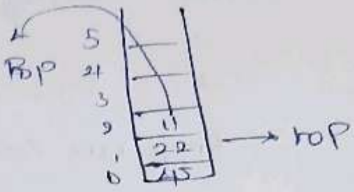
x. pop( )



top = 3-1

top = 2

x. pop( )

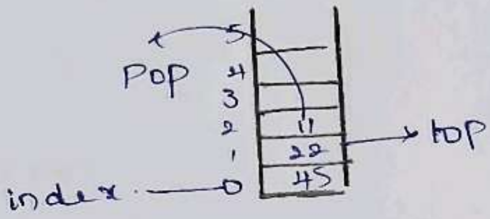


top = 2-1

top = 1

∴ Peak element = 22

index position = 1



10

Perform the following operations in Circular Queue having a size of 5.

size = 5

Enqueue (20)

Enqueue (10)

Enqueue (30)

Dequeue ( )

Enqueue (80)

Dequeue ( ),

Enqueue (90)

Enqueue (100)

Dequeue ( ),

Dequeue (60),

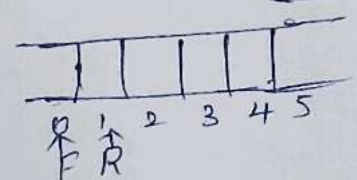
Enqueue (90).

size = 5

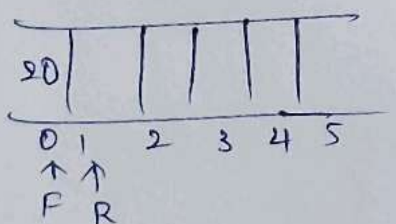
Rare = Front + 1

Front = Rare - 1

enter.

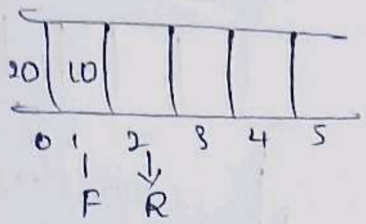


(i) Enqueue (20)



FRont = Rare - 1
= 1 - 1 = 0.

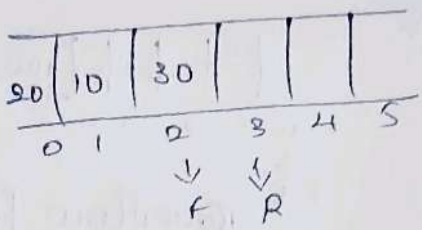
(ii) Enqueue (10)



FRont = 2 - 1

FRont = 1

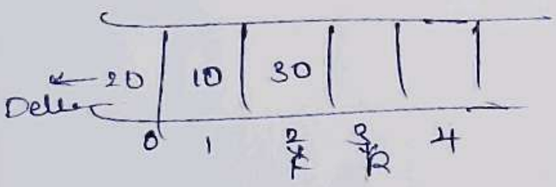
(iii) Enqueue (30)



Front = 3 - 1

Front = 2

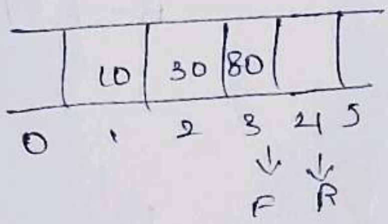
(iv) Dequeue ( )



Front = 3 - 1

Front = 2

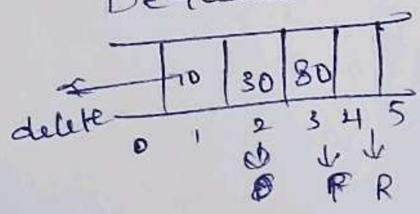
(v) Enqueue (80)



FRont = 4 - 1

FRont = 3

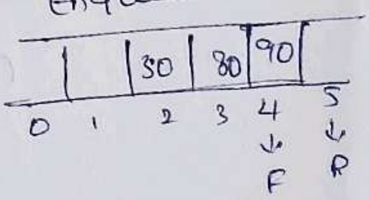
(vi) Dequeue ( )



FRont = 4 - 1

FRont = 3

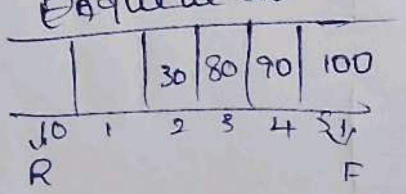
(vii) Enqueue (90)



Front = 5 - 1

Front = 4

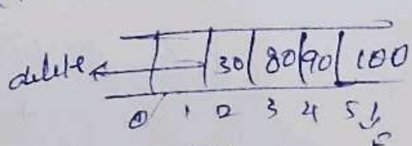
(viii) Enqueue (100)



FRont = 0 - ~~1~~

= 5

(ix) Deque ( )



FRont = 0 - 1

= 5 .

(xii) Dequeue (60)

			80	90	100
0	1	2	3	4	5

FR

overflow $\boxed{top = N-1}$

Error

The queue is full

(xiii) Dequeue (90)

			80	90	100
0	1	2	3	4	5

FR

overflow.

3. Convert the following infix expression into a post expression.

$$B - (C/D + (E \% F * G) / H) * J$$

B	B	B
-	-	B·
(	C -	
C	C -	
/	C -	
D	C -	
+	C -	

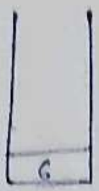
C	~~C C -~~	
E	~~C C -~~	
B	B	B
-	-	B
(	- C	B
C	- C	BC
/	- C /	BC
D	- C /	BCD.
+	- C +	BCD /
(	- C + C	BCD /
E	- C + C	BCD/E
%	- C + C %	BCD/E.

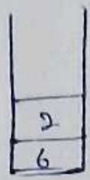
F	% C + C -	BCD/EF
*	* C + C -	BCD/EF%
G	* C + C -	BCD/EF%G
)	) * C + C -	BCD/EF%G-
/	/ C + -	BCD/EF%G•*
H	/ C + -	BCD/EF%G* H
)	C • -	BCD/EF%G*H/+
J	-	BCD/EF%G*H/+J
over		BCD/EF%G*H/+J* (error)

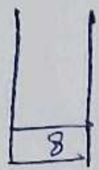
(a) 9 3 8 4 / * +

Symbol	Stack (top → bottom)	
9	9	
3	3, 9	
8	8, 3, 9	
4	4, 8, 3, 9	
/	2, 3, 9	$8/4 = 2$
*	6, 9	$3 * 2 = 6$
+	15	$9 + 6 = 15$
over		O/P = 15.

(b) 6 2 + 6 8 / *

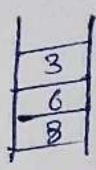
6 

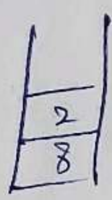
2 

+ 

$6+2=8$

~~8~~ 6 

3 

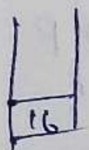
/ 

$3/6 = 2$

* 

$8 * 2 = 16$

8

over 

O/P : 16.

Annexure A – Analytical Problem Solving

1) Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(),Pop(). Print the peak element of the data structure and its index position. With a neat sketch, explain in detail.

2)Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail.

3)Convert the following infix expression into a post fix expression.
B - (C/D + (E % F* G) / H) * J.
Also, evaluate the given postfix expressions. a) 9 3 8 4 /*+ b) 6 2 + 6 3 /*

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _______________	Signature: _______________	Signature: _______________
Date:	Date:	Date: